

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407405
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Macias; John F.

Systems and methods for conditional handover using non-terrestrial node telemetry information

Abstract

A device described herein, such as a User Equipment (“UE”), may connect to a base station of a wireless network, receive telemetry information associated with a plurality of Non-Terrestrial Network (“NTN”) nodes, identify a particular handover condition, select, based on identifying the particular handover condition, a particular NTN node of the plurality of NTN nodes, initiate a handover procedure from the first base station of the wireless network to the selected particular NTN node, and connect to the particular NTN node based on the handover procedure. The NTN nodes may be implemented by satellites that orbit the Earth.

Inventors:	Macias; John F. (Antelope, CA)
Applicant:	Verizon Patent and Licensing Inc. (Basking Ridge, NJ)
Family ID:	91472364
Assignee:	Verizon Patent and Licensing Inc. (Basking Ridge, NJ)
Appl. No.:	18/068052
Filed:	December 19, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240204862 A1	Jun. 20, 2024

Publication Classification

Int. Cl.: H04B7/185 (20060101); H04W36/08 (20090101); H04W36/24 (20090101)

U.S. Cl.:

CPC **H04B7/18513** (20130101); **H04W36/08** (20130101); **H04W36/083** (20230501);
H04W36/24 (20130101);

Field of Classification Search

CPC: H04B (7/18513); H04B (7/18541); H04W (36/08); H04W (36/083); H04W (36/24);
 H04W (36/322); H04W (36/085); H04W (36/087); H04W (36/13); H04W (36/247);
 H04W (36/249)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10805850	12/2019	Benammar	N/A	H04W 36/144
2023/0041601	12/2022	Vangala	N/A	H04W 36/328
2024/0244517	12/2023	Lee	N/A	H04W 48/20

OTHER PUBLICATIONS

Third Generation Partnership Project, “Technical Specification Group Services and System Aspects; Study on using Satellite Access in 5G; Stage 1 (Release 16),” TR 22.822 V16.0.0 (Jun. 2018). cited by applicant

Third Generation Partnership Project, “Technical Specification Group Radio Access Network; NR; NR and NG-RAN Overall Description; Stage 2 (Release 17),” 3GPP TS 38.300 V17.2.0 (Sep. 2022). cited by applicant

“Technical Specification Group Radio Access Network; Study on Narrow-Band Internet of Things (NB-IoT) / enhanced Machine Type Communication (eMTC) support for Non-Terrestrial Networks (NTN) (Release 17)”, Third Generation Partnership Project, 3GPP TR 36.763 V17.0.0 (Jun. 2021). cited by applicant

Primary Examiner: Yea; Ji-Hae

Background/Summary

BACKGROUND

(1) Wireless networks provide wireless connectivity to user equipment (“UE”), such as mobile telephones or other wireless communication devices. Wireless networks may include wireless network infrastructure, such as ground-based base stations, which may be installed on buildings, towers, or other structures. Some wireless networks may include Non-Terrestrial Networks (“NTNs”), which may utilize satellites or other machines, equipment, etc. that orbit the Earth and which may not have a fixed position relative to the ground.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example of a group of NTN nodes;
- (2) FIGS. 2 and 3 illustrate an example of telemetry information, associated with respective NTN

nodes, that the NTN nodes may each maintain;

(3) FIG. 4 illustrates an example of telemetry information, associated with multiple NTN nodes of a group, that each NTN node of the group may maintain;

(4) FIG. 5A illustrates an example of a UE receiving a set of handover conditions and NTN group telemetry information from an NTN node after connecting to the NTN node, in accordance with some embodiments;

(5) FIG. 5B illustrates an example of a UE receiving a set of handover conditions and NTN group telemetry information from an NTN node prior to connecting to the NTN node, in accordance with some embodiments;

(6) FIG. 6 illustrates an example of a UE detecting a handover condition based on conditional handover information and/or NTN group telemetry information, in accordance with some embodiments;

(7) FIG. 7 illustrates an example of a UE connected to a second NTN node after initiating a handover from a first NTN node, in accordance with some embodiments;

(8) FIG. 8 illustrates an example of a UE detecting a handover condition while connected to a ground base station, and selecting a particular NTN node based on NTN group telemetry information, in accordance with some embodiments;

(9) FIG. 9 illustrates an example of a UE connected to an NTN node after initiating a handover from a ground base station, in accordance with some embodiments;

(10) FIG. 10 illustrates an example process for performing a UE-initiated handover based on NTN node telemetry information, in accordance with some embodiments;

(11) FIG. 11 illustrates an example environment in which one or more embodiments, described herein, may be implemented;

(12) FIG. 12 illustrates an example arrangement of a radio access network (“RAN”), in accordance with some embodiments; and

(13) FIG. 13 illustrates example components of one or more devices, in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

(14) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

(15) Embodiments described herein provide for a UE to detect handover conditions and initiate handovers to and/or from NTN nodes, such as satellites or other wireless network infrastructure equipment that orbits the Earth and/or is otherwise mobile and/or airborne. For example, as discussed below, a UE may receive NTN telemetry information, based on which the UE may select a particular NTN node, and may initiate a handover to the selected particular NTN node. In this manner, the wireless coverage provided by a wireless network may be enhanced to include connectivity to NTN nodes. Further, since embodiments described herein afford UEs the opportunity to detect handover conditions (e.g., based on UE-detected signal strength metrics, UE-determined latency or other performance metrics, UE-determined application or traffic types, etc.), the options available to UEs to initiate handovers (e.g., handover targets) may be widened by the availability of NTN nodes, thus enhancing the coverage of wireless services offered to UEs.

(16) As shown in FIG. 1, multiple NTN nodes **101** may orbit the Earth and/or may otherwise be mobile and/or airborne. As discussed above, NTN nodes **101** may include and/or may be implemented by satellites and/or other suitable apparatus that includes wireless network infrastructure equipment (e.g., radios, antennas, etc.). NTN nodes **101** may implement, for example, a base station of RAN, a Distributed Unit (“DU”) of a RAN, and/or some other element of a wireless network that provides a wireless interface between UEs, such as mobile phones, Internet of Things (“IoT”) devices, etc. and a core network (e.g., an Evolved Packet Core (“EPC”), a Fifth Generation (“5G”) core (“5GC”), etc.). In the examples described herein, each NTN node **101** is described as being implemented by a single satellite or other suitable apparatus. However,

concepts similar to those described herein apply to other arrangements or implementations, such as implementations in which a single satellite or other suitable apparatus implements multiple NTN nodes **101**.

(17) As further shown in FIG. **1**, a set of NTN nodes may be associated with a particular NTN node group **103**. For example, NTN node group **103** may include NTN nodes **101** that are configured to communicate with each other and/or otherwise to receive information associated with each other, such as telemetry information as discussed below. In some implementations, NTN node group **103** may be referred to as, or implemented by, a “constellation” of satellites. In some embodiments, NTN node group **103** may not be a predetermined group, but may instead refer to a set of NTN nodes **101** that are within a threshold proximity of each other (e.g., within wireless communication range of each other). For example, NTN nodes **101** of NTN node group **103** may be “neighbors” of each other, inasmuch as NTN nodes **101** of NTN node group **103** may be located physically closer to each other than other NTN nodes **101** (e.g., other NTN nodes **101** that are orbiting the Earth).

(18) Each NTN node **101** may generate, receive, or otherwise maintain a set of telemetry information, “ephemeris” information, etc. (referred to herein simply as “telemetry information” for the sake of brevity). For example, as shown in FIG. **2**, NTN node **101-1** may maintain a respective set of NTN node telemetry information **201-1**. NTN node telemetry information **201-1** may include, for example, a current position of NTN node **101-1** and a trajectory of NTN node **101-1**. The current position may be expressed in terms of latitude and longitude coordinates, altitude, and/or other suitable position information indicating where NTN node **101-1** is in relation to a reference point or line on the Earth (e.g., the North Pole, the Equator, the Prime Meridian, sea level, etc.). The trajectory may be expressed in terms of heading, speed, target location(s) and predicted or estimated time of arrival, a flight plan, and/or other suitable information based on which a future location of NTN node **101-1** may be determined. In some embodiments, NTN node telemetry information **201-1** may include other suitable information describing telemetry of NTN node **101-1**, such as angle, acceleration, or other suitable information. NTN node telemetry information **201-1** may be determined by NTN node **101-1** based on one or more onboard sensors, and/or may be received from some other source, such as a terrestrial-based control station, another NTN node **101**, and/or some other suitable device or system. NTN node telemetry information **201-1** may also include a timestamp, indicating a time at which the position, trajectory, etc. have been determined.

(19) In this manner, as shown in FIG. **3**, each NTN node **101** (e.g., NTN nodes **101-1**, **101-2**, and **101-3**) of NTN node group **103** may maintain its own respective NTN node telemetry information **201** (e.g., NTN node **101-2** may maintain NTN node telemetry information **201-2** and NTN node **101-3** may maintain NTN node telemetry information **201-3**). Additionally, as shown in FIG. **4**, each NTN node **101** of NTN node group **103** may maintain NTN node telemetry information **201** of some or all of the other NTN nodes **101** of NTN node group **103**. In the example shown, NTN nodes **101-1**, **101-2**, and **101-3** may maintain a set of group telemetry information **401** (e.g., an “almanac”), which may include the respective NTN node telemetry information **201** of each NTN node **101** of NTN node group **103** (e.g., NTN node telemetry information **201-1**, **201-2**, and **201-3**, in this example). For example, in addition to maintaining its own NTN node telemetry information **201-1**, NTN node **101-1** may also maintain NTN node telemetry information **201-2** and **201-3** associated with NTN nodes **101-2** and **101-3**, respectively. Similarly, in addition to maintaining its own NTN node telemetry information **201-2**, NTN node **101-2** may also maintain NTN node telemetry information **201-1** and **201-3** associated with NTN nodes **101-1** and **101-3**, respectively.

(20) NTN nodes **101** may receive NTN node telemetry information **201** associated with other NTN nodes **101** by wirelessly communicating directly with such NTN nodes **101**. For example, NTN nodes **101-1** and **101-2** may wirelessly communicate with each other in order for NTN node **101-1** to provide NTN node telemetry information **201-1** to NTN node **101-2**, and for NTN node **101-2** to provide NTN node telemetry information **201-2** to NTN node **101-1**. Additionally, or alternatively,

NTN nodes **101** may communicate indirectly via one or more relays, such as ground-based devices or systems and/or via one or more other airborne devices or systems. For example, NTN node **101-1** may provide NTN node telemetry information **201-1** to such relay device or system, which may forward NTN node telemetry information **201-1** to NTN nodes **101-2** and/or **101-3**. In some situations, group telemetry information **401** may include different versions of NTN node telemetry information **201** associated with respective NTN nodes **101**. For example, NTN node telemetry information **201-2**, maintained by NTN node **101-1**, may be less up-to-date than NTN node telemetry information **201-2** as maintained by NTN node **101-2**. For example, NTN node **101-2** may update a local copy of NTN node telemetry information **201-2** more frequently than NTN node **101-1** receives updated NTN node telemetry information **201-2**.

(21) As discussed herein, group telemetry information **401** and/or NTN node telemetry information **201** associated with one or more NTN nodes **101** may be used by a UE when detecting handover conditions, selecting handover targets, and initiating handovers to such handover targets. For example, as shown in FIG. 5A, UE **501** may scan for and connect (at **502**) to NTN node **101-1**. UE **501** may be an NTN-capable UE, inasmuch as one or more radios, antennas, etc. of UE **501** may operate according to the same frequency, frequency band, radio access technology (“RAT”), etc. implemented by NTN node **101-1**.

(22) In some implementations, UE **501** may scan for and connect (at **502**) to NTN node **101-1** based on the absence of a ground-based base station. For example, prior to scanning (at **502**) for NTN node **101-1** (e.g., attempting to locate wireless signals such as Master Information Blocks (“MIBs”), System Information Blocks (“SIBs”), timing and/or synchronization signals, presence signals, etc. outputted by one or more NTN nodes **101**), UE **501** may have scanned for one or more ground-based base stations according to a frequency, frequency band, RAT, etc. implemented by such ground-based base stations. In such a situation, UE **501** may have not detected the presence of a ground-based base station, based on which UE **501** may scan (at **502**) according to a frequency, frequency band, RAT, etc. associated with NTN nodes **101**.

(23) Connecting to NTN node **101-1** may include performing a Radio Resource Control (“RRC”) connection setup, exchanging one or more RRC messages, etc. As part of, in conjunction with, or subsequent to the respective connection setup, NTN node **101** may request attributes, capabilities, etc. of UE **501**. UE **501** may respond with an indication that UE **501** supports conditional handover procedures, and/or may respond with a request for conditional handover information.

(24) In response to the indication that UE **501** supports conditional handover procedures, in response to the request for conditional handover information, and/or based on one or more other factors (e.g., based on the connection of UE **501** to NTN node **101-1**), NTN node **101-1** may respond with a set of handover conditions **503**. NTN handover conditions **503** may specify conditions based on which UE **501** may request a handover from NTN node **101-1** (or other NTN nodes **101** and/or wireless network infrastructure, including ground-based base stations) to another NTN node **101** and/or to some other wireless network infrastructure. NTN node **101-1** may, in some embodiments, receive NTN handover conditions **503** from a core network or some other suitable source.

(25) As one example, NTN handover conditions **503** may specify that UE **501** should request a handover from a particular NTN node **101**, to which UE **501** is connected, to another NTN node **101** if signal strength metrics between UE **501** and the particular NTN node **101** (e.g., as measured by UE **501**) are below a first threshold and if signal strength metrics between UE **501** and the other NTN node **101** are above a second threshold. As another example, NTN handover conditions **503** may specify that UE **501** should request a handover from the particular NTN node **101**, to which UE **501** is connected, to another NTN node **101** if a distance between UE **501** and the particular NTN node **101** (e.g., as determined by UE **501** based on receiving group telemetry information **401**) is greater than a first threshold distance and if distance metrics between UE **501** and the other NTN node **101** is less than a second threshold distance.

(26) As shown, NTN node **101-1** may also provide (at **504**) group telemetry information **401**, which may include NTN node telemetry information **201-1** associated with NTN node **101-1** as well as NTN node telemetry information **201** associated with one or more other NTN nodes **101** (e.g., NTN nodes **101** of the same NTN node group **103** as NTN node **101-1**). In some embodiments, UE **501** may receive (at **504**) group telemetry information **401** as part of, in conjunction with, or subsequent to the connection (at **502**) of UE **501** to NTN node **101-1**. In some embodiments, UE **501** may additionally receive, on an ongoing basis and while UE **501** is connected to NTN node **101-1**, updated group telemetry information **401** (e.g., in the event that NTN node **101-1** receives updates to some or all of group telemetry information **401** over time, such as course changes or updated positions associated with NTN node **101-1** and/or one or more other NTN nodes **101**). In this manner, UE **501** may be “aware” of the positions of NTN node **101-1** and one or more other NTN nodes **101** at any given time, including future times (e.g., based on a present position and trajectory of respective NTN nodes **101**).

(27) In some embodiments, UE **501** may receive group telemetry information **401** and/or NTN handover conditions **503** without connecting to a given NTN node **101**. For example, as shown in FIG. 5B, UE **501** may receive (at **506**) NTN handover conditions **503** and/or group telemetry information **401** from a source other than NTN node **101-1**, and/or from NTN node **101-1** prior to connecting to NTN node **101-1**. For example, NTN node **101-1**, a ground-based base station, and/or some other device or system may provide (at **506**) NTN handover conditions **503** and group telemetry information **401** prior to UE **501** scanning for and connecting to (at **508**) NTN node **101-1**. In some embodiments, NTN node **101-1**, another NTN node **101**, and/or some other device or system may broadcast NTN handover conditions **503** and/or group telemetry information **401** as part of a MIB, SIB, or other type of control signaling. Additionally, or alternatively, NTN node **101-1**, another NTN node **101**, and/or some other device or system may broadcast NTN handover conditions **503** and/or group telemetry information **401** using a Multimedia Broadcast Multicast Service (“MBMS”) technique, an evolved MBMS (“eMBMS”) technique, and/or some other suitable type of broadcast technique. Additionally, or alternatively, UE **501** may have received (at **506**) NTN handover conditions **503** and/or group telemetry information **401** during an initial provisioning procedure, an Over-the-Air (“OTA”) update procedure, and/or via some other suitable procedure. Accordingly, based on the techniques described in FIG. 5A or 5B, and/or some other suitable technique, UE **501** may receive and maintain a set of NTN handover conditions **503** as well as group telemetry information **401** associated with one or more NTN nodes **101** (e.g., associated with one or more NTN node groups **103**).

(28) FIGS. 6 and 7 illustrate an example handover that may be initiated by a given UE **501** when such UE **501** is currently connected to a given NTN node **101**. For example, as shown in FIG. 6, UE **501** may be connected to NTN node **101-1**. At some point, while UE **501** is connected to NTN node **101-1**, UE **501** may detect (at **602**) a handover condition. For example, UE **501** may compare group telemetry information **401** (e.g., NTN node telemetry information **201-1** associated with NTN node **101-1** and/or NTN node telemetry information **201-2** associated with NTN node **101-2**), measures of signal strength or quality associated with NTN nodes **101-1** and/or **101-2**, and/or other factors, to one or more NTN handover conditions **503**.

(29) In this example, UE **501** may identify, based on group telemetry information **401**, a trajectory, course, heading, direction, etc. (hereinafter referred to as trajectory **601**) associated with each NTN node **101**. For example, UE **501** may identify trajectory **601-1** associated with NTN node **101-1**, and may identify trajectory **601-2** associated with NTN node **101-2**. Such trajectories **601** may indicate, in this example, that NTN node **101-1** is moving away from UE **501** (e.g., a distance between NTN node **101-1** and UE **501** may be increasing as a function of time) and that NTN node **101-2** is moving toward UE **501** (e.g., a distance between NTN node **101-2** and UE **501** may be decreasing as a function of time). In some embodiments, UE **501** may identify one or more other metrics, Key Performance Indicators (“KPIs”), events, etc. For example, UE **501** may identify that

a measure of signal strength or quality (e.g., Signal-to-Interference-and-Noise-Ratio (“SINR”), Reference Signal Received Power (“RSRP”), Reference Signal Received Quality (“RSRQ”), etc.) between UE **501** and NTN node **101-1** is decreasing and/or is below a first threshold, and/or may identify that a measure of signal strength or quality between UE **501** and NTN node **101-2** is increasing and/or is above a second threshold.

(30) Based on identifying the relative trajectories **601** of NTN nodes **101-1** and **101-2** and/or based on other factors, UE **501** may determine that a particular handover condition, associated with NTN handover conditions **503**, has been met. UE **501** may further identify that NTN node **101-2** is a suitable handover target based on criteria, instructions, or actions associated with the particular determined handover condition. In some embodiments, UE **501** may identify that NTN node **101-2** is a suitable handover target based on signal strength or quality measurements performed by UE **501** (e.g., NTN node **101-2** may be a NTN node **101** with the highest measure of signal strength or quality with which UE **501** is able to wirelessly communicate). In some embodiments, UE **501** may identify that NTN node **101-2** is a suitable handover target based on one or more other techniques.

(31) UE **501** may accordingly initiate (at **604**) a handover to NTN node **101-2**. For example, UE **501** may output a handover request to NTN node **101-1** and/or to NTN node **101-2**, indicating NTN node **101-2** as a handover target and/or indicating NTN node **101-1** as a handover source. NTN nodes **101-1** and **101-2** may accordingly communicate with each other, with UE **501**, and/or one or more other devices or systems (e.g., an Access and Mobility Management Function (“AMF”), a Mobility Management Entity (“MME”), and/or some other network function) in order to effectuate the UE-initiated handover of UE **501** from NTN node **101-1** to **101-2**.

(32) After the handover has been performed, and as shown in FIG. 7, UE **501** may be connected to NTN node **101-2** and may no longer be connected to NTN node **101-1**. In some embodiments, the result of the handover may be different than the example shown in FIG. 7. For example, the handover may result in a connection between UE **501** and NTN node **101-2** being established, and a connection between UE **501** and NTN node **101-1** remaining established and/or being set to a different mode or state (e.g., an idle mode, a secondary mode, etc.).

(33) FIGS. 7 and 8 illustrate another example handover that may be initiated by UE **501** based on NTN node telemetry information **201** and/or group telemetry information **401** associated with one or more NTN nodes **101** and/or NTN node groups **103**. As shown, UE **501** may be connected (at **802**) to ground-based base station (also referred to as “ground base station”) **801**. For example, as discussed above, ground base station **801** may be mounted, installed, affixed, etc. to a building, a tower, a vehicle, and/or some other ground-based device or structure. UE **501** may further receive (at **804**) group telemetry information **401** (e.g., NTN node telemetry information **201** associated with one or more NTN nodes **101**, such as NTN nodes **101-1** and/or **101-2**) and/or may receive a set of NTN handover conditions **503**. For example, as discussed above with respect to FIG. 5 (at **506**), UE **501** may receive group telemetry information **401** and/or NTN handover conditions **503** from one or more NTN nodes **101** (e.g., which may broadcast such information), from ground base station **801** (e.g., which may serve as a relay for such information, which may be received from NTN nodes **101** and/or from some other device or system that provides such information), and/or from some other source. For example, UE **501** may receive group telemetry information **401** on an ongoing basis, such as when NTN node telemetry information **201** associated with one or more NTN nodes **101** is updated or otherwise provided.

(34) At some point, UE **501** may determine (at **806**) a handover condition, such as detecting the UE **501** is moving away from ground base station **801**, that a signal strength between UE **501** and ground base station **801** is approaching or is below a threshold measure of signal strength, that UE **501** is initiating a particular application or service type indicated in a particular set of handover conditions **503** (e.g., is initiating an emergency call, is initiating a secure message session, etc.), and/or may identify the handover condition based on one or more other UE-determined metrics or events.

(35) UE **501** may select (at **808**) NTN node **101-2** as a handover target based on determining (at **806**) the handover condition. For example, UE **501** may utilize group telemetry information **401** (e.g., NTN node telemetry information **201-1** associated with NTN node **101-1** and/or NTN node telemetry information **201-2** associated with NTN node **101-2**) to select (at **808**) NTN node **101-2** as a handover target for UE **501**. Based on group telemetry information **401**, UE **501** may determine that NTN node **101-2** is closer in position to UE **501** than one or more other NTN nodes **101** (e.g., NTN node **101-1**). As another example, UE **501** may receive or maintain information (e.g., from a user of UE **501**) indicating a predicted or expected route (e.g., a navigation route of UE **501** from one geographical location to another), and may further identify that a trajectory of NTN node **101-2** matches the predicted or expected route of UE **501** most closely out of a set of NTN nodes **101**. That is, UE **501** may select (at **808**) NTN node **101-2** based on factors in addition to, or in lieu of, signal strength or quality metrics between UE **501** and NTN node **101-2**, such as based on a trajectory of NTN node **101-2** and/or one or more other NTN nodes **101**. Thus, as shown in FIG. **9**, UE **501** may initiate (at **910**) a handover from ground base station **801** to NTN node **101-2**, resulting in a connection being established between UE **501** and NTN node **101-2**, and a connection between UE **501** being de-established or otherwise modified based on the handover.

(36) FIG. **10** illustrates an example process **1000** for performing a UE-initiated handover based on NTN node telemetry information. In some embodiments, some or all of process **1000** may be performed by UE **501**. In some embodiments, one or more other devices may perform some or all of process **1000** in concert with, and/or in lieu of, UE **501**.

(37) As shown, process **1000** may include connecting (at **1002**) to a base station of a wireless network. For example, UE **501** may connection to a base station implemented by a satellite (e.g., may connect to a particular NTN node **101**) or to a ground-based base station (e.g., ground base station **801**).

(38) Process **1000** may further include receiving (at **1004**) handover condition information. For example, UE **501** may receive handover conditions **503** from the base station to which UE **501** is currently connected (e.g., from NTN node **101**, from ground base station **801**, etc.) and/or may receive handover conditions **503** from some other source (e.g., via a broadcast or some other suitable technique). As discussed above, handover conditions **503** may include factors, criteria, conditions, thresholds, etc. based on information that may be measured, collected, etc. by UE **501**. Such factors, criteria, etc. may be based on signal strength metrics measured by UE **501**, a predicted or expected location of UE **501**, and/or other UE-determined metrics or events. As another example, handover conditions **503** may include factors, criteria, etc. associated with NTN nodes **101**, such as relative location of a given NTN node **101** to UE **501** and/or a trajectory of NTN node **101** relative to UE **501**. For example, NTN handover conditions **503** may include conditions relating to whether NTN node **101** is moving towards UE **501**, is moving away from UE **501**, is within a threshold proximity of UE **501** for at least a threshold amount of time based on a trajectory of NTN node **101** and a predicted or estimated path of UE **501**, and/or other NTN node **101** trajectory-based conditions.

(39) Process **1000** may additionally include receiving and/or monitoring (at **1006**) telemetry information associated with one or more NTN nodes **101**. For example, NTN nodes **101** may maintain their own respective NTN node telemetry information **201** (e.g., “ephemeris” data) as well as NTN node telemetry information **201** associated with one or more other NTN nodes **101** (e.g., an “almanac”), such that one particular NTN node **101** may provide NTN node telemetry information **201** associated with multiple NTN nodes **101** (e.g., group telemetry information **401**). As discussed above, UE **501** may receive NTN node telemetry information **201** and/or group telemetry information **401** from a particular NTN node **101** to which UE **501** is currently connected, may receive NTN node telemetry information **201** and/or group telemetry information **401** from a particular NTN node **101** to which UE **501** is not currently connected (e.g., via one or more broadcasts by the particular NTN node **101**), one or more relays, and/or from some other

suitable source. UE **501** may receive NTN node telemetry information **201** and/or group telemetry information **401** on an ongoing basis, such as when such information changes or when other NTN node telemetry information **201** associated with one or more other NTN nodes **101** becomes available (e.g., such NTN nodes **101** may move towards another NTN node **101** that is currently providing NTN node telemetry information **201** and/or group telemetry information **401** to UE **501**).

(40) Process **1000** may also include identifying (at **1008**) a handover condition. For example, UE **501** may compare UE-determined metrics or information, NTN node telemetry information **201** associated with one or more NTN nodes **101**, and/or other suitable information to handover conditions **503** and may identify that at least a particular handover condition is satisfied.

(41) Process **1000** may further include selecting (at **1010**) a particular NTN node **101** as a handover target. For example, UE **501** may select the particular NTN node **101**, from a group of NTN nodes **101** for which UE **501** has received NTN node telemetry information **201**, based on a trajectory of the particular NTN node **101**, a location of NTN node **101**, and/or one or more other factors. In some embodiments, the selected NTN node **101** may be a different NTN node **101** from which UE **501** received NTN node telemetry information **201** and/or group telemetry information **401**.

Generally, UE **501** may identify that the particular NTN node **101** is the closest NTN node **101** to UE **501**, is predicted or estimated to be the closest NTN node **101** or within a particular distance of UE **501** for at least a threshold amount of time (e.g., based on a location of UE **501** and/or based on a predicted or estimated path or route of UE **501**), and/or is otherwise a “best match” for a handover target for UE **501** based on NTN node telemetry information **201** associated with the particular NTN node **101** and/or other NTN nodes **101**.

(42) Process **1000** may additionally include initiating (at **1012**) a handover to the selected NTN node **101**. For example, as discussed above, UE **501** may output one or more messages to the base station to which UE **501** is currently connected, and/or may output one or more messages to the selected NTN node **101**, to initiate the handover. The currently connected base station, the selected NTN node **101**, and/or one or more other devices or systems may communicate in order to effectuate the handover, after which UE **501** may communicate with the wireless network via the particular NTN node **101** (e.g., may be handed over from the source base station, to which UE **501** was previously connected, to the selected NTN node **101**).

(43) FIG. **11** illustrates an example environment **1100**, in which one or more embodiments may be implemented. In some embodiments, environment **1100** may correspond to a Fifth Generation (“5G”) network, and/or may include elements of a 5G network. In some embodiments, environment **1100** may correspond to a 5G Non-Standalone (“NSA”) architecture, in which a 5G radio access technology (“RAT”) may be used in conjunction with one or more other RATs (e.g., a Long-Term Evolution (“LTE”) RAT), and/or in which elements of a 5G core network may be implemented by, may be communicatively coupled with, and/or may include elements of another type of core network (e.g., an evolved packet core (“EPC”). In some embodiments, portions of environment **1100** may represent or may include a 5G core (“5GC”). As shown, environment **1100** may include UE **501**, RAN **1110** (which may include one or more Next Generation Node Bs (“gNBs”) **1111**), RAN **1112** (which may include one or more evolved Node Bs (“eNBs”) **1113**), and various network functions such as AMF **1115**, MME **1116**, Serving Gateway (“SGW”) **1117**, Session Management Function (“SMF”)/Packet Data Network (“PDN”) Gateway (“PGW”)-Control plane function (“PGW-C”) **1120**, Policy Control Function (“PCF”)/Policy Charging and Rules Function (“PCRF”) **1125**, Application Function (“AF”) **1130**, User Plane Function (“UPF”)/PGW-User plane function (“PGW-U”) **1135**, Unified Data Management (“UDM”)/Home Subscriber Server (“HSS”) **1140**, and Authentication Server Function (“AUSF”) **1145**.

Environment **1100** may also include one or more networks, such as Data Network (“DN”) **1150**.

(44) The example shown in FIG. **11** illustrates one instance of each network component or function (e.g., one instance of SMF/PGW-C **1120**, PCF/PCRF **1125**, UPF/PGW-U **1135**, UDM/HSS **1140**,

and/or AUSF **1145**). In practice, environment **1100** may include multiple instances of such components or functions. For example, in some embodiments, environment **1100** may include multiple “slices” of a core network, where each slice includes a discrete and/or logical set of network functions (e.g., one slice may include a first instance of SMF/PGW-C **1120**, PCF/PCRF **1125**, UPF/PGW-U **1135**, UDM/HSS **1140**, and/or AUSF **1145**, while another slice may include a second instance of SMF/PGW-C **1120**, PCF/PCRF **1125**, UPF/PGW-U **1135**, UDM/HSS **1140**, and/or AUSF **1145**). The different slices may provide differentiated levels of service, such as service in accordance with different Quality of Service (“QoS”) parameters.

(45) The quantity of devices and/or networks, illustrated in FIG. **11**, is provided for explanatory purposes only. In practice, environment **1100** may include additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than illustrated in FIG. **11**. For example, while not shown, environment **1100** may include devices that facilitate or enable communication between various components shown in environment **1100**, such as routers, modems, gateways, switches, hubs, etc. In some implementations, one or more devices of environment **1100** may be physically integrated in, and/or may be physically attached to, one or more other devices of environment **1100**. Alternatively, or additionally, one or more of the devices of environment **1100** may perform one or more network functions described as being performed by another one or more of the devices of environment **1100**.

(46) Elements of environment **1100** may interconnect with each other and/or other devices via wired connections, wireless connections, or a combination of wired and wireless connections. Examples of interfaces or communication pathways between the elements of environment **1100**, as shown in FIG. **11**, may include an N1 interface, an N2 interface, an N3 interface, an N4 interface, an N5 interface, an N6 interface, an N7 interface, an N8 interface, an N9 interface, an N10 interface, an N11 interface, an N12 interface, an N13 interface, an N14 interface, an N2 interface, an N26 interface, an S1-C interface, an S1-U interface, an S5-C interface, an S5-U interface, an S6a interface, an S11 interface, and/or one or more other interfaces. Such interfaces may include interfaces not explicitly shown in FIG. **11**, such as Service-Based Interfaces (“SBIs”), including an Namf interface, an Nudm interface, an Npcf interface, an Nupf interface, an Nnef interface, an Nsmf interface, and/or one or more other SBIs.

(47) UE **501** may include a computation and communication device, such as a wireless mobile communication device that is capable of communicating with RAN **1110**, RAN **1112**, and/or DN **1150**. UE **501** may be, or may include, a radiotelephone, a personal communications system (“PCS”) terminal (e.g., a device that combines a cellular radiotelephone with data processing and data communications capabilities), a personal digital assistant (“PDA”) (e.g., a device that may include a radiotelephone, a pager, Internet/intranet access, etc.), a smart phone, a laptop computer, a tablet computer, a camera, a personal gaming system, an Internet of Things (“IoT”) device (e.g., a sensor, a smart home appliance, a wearable device, a Machine-to-Machine (“M2M”) device, or the like), or another type of mobile computation and communication device. UE **501** may send traffic to and/or receive traffic (e.g., user plane traffic) from DN **1150** via RAN **1110**, RAN **1112**, and/or UPF/PGW-U **1135**.

(48) RAN **1110** may be, or may include, a 5G RAN that includes one or more base stations (e.g., one or more gNBs **1111**), via which UE **501** may communicate with one or more other elements of environment **1100**. UE **501** may communicate with RAN **1110** via an air interface (e.g., as provided by gNB **1111**). For instance, RAN **1110** may receive traffic (e.g., user plane traffic such as voice call traffic, data traffic, messaging traffic, etc.) from UE **501** via the air interface, and may communicate the traffic to UPF/PGW-U **1135** and/or one or more other devices or networks. Further, RAN **1110** may receive signaling traffic, control plane traffic, etc. from UE **501** via the air interface, and may communicate such signaling traffic, control plane traffic, etc. to AMF **1115** and/or one or more other devices or networks. Additionally, RAN **1110** may receive traffic intended

for UE **501** (e.g., from UPF/PGW-U **1135**, AMF **1115**, and/or one or more other devices or networks) and may communicate the traffic to UE **501** via the air interface. In some embodiments, ground base station **801** may be, may include, and/or may be implemented by one or more gNBs **1111**. In some embodiments, one or more NTN nodes **101** may be, may include, may implement, etc. one or more gNBs **1111**. As such, RAN **1110** may include one or more ground base stations **801** and/or one or more NTN nodes **101**.

(49) RAN **1112** may be, or may include, a LTE RAN that includes one or more base stations (e.g., one or more eNBs **1113**), via which UE **501** may communicate with one or more other elements of environment **1100**. UE **501** may communicate with RAN **1112** via an air interface (e.g., as provided by eNB **1113**). For instance, RAN **1112** may receive traffic (e.g., user plane traffic such as voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE **501** via the air interface, and may communicate the traffic to UPF/PGW-U **1135** (e.g., via SGW **1117**) and/or one or more other devices or networks. Further, RAN **1112** may receive signaling traffic, control plane traffic, etc. from UE **501** via the air interface, and may communicate such signaling traffic, control plane traffic, etc. to MME **1116** and/or one or more other devices or networks. Additionally, RAN **1112** may receive traffic intended for UE **501** (e.g., from UPF/PGW-U **1135**, MME **1116**, SGW **1117**, and/or one or more other devices or networks) and may communicate the traffic to UE **501** via the air interface. In some embodiments, ground base station **801** may be, may include, and/or may be implemented by one or more eNBs **1113**. In some embodiments, one or more NTN nodes **101** may be, may include, may implement, etc. one or more eNBs **1113**. As such, RAN **1112** may include one or more ground base stations **801** and/or one or more NTN nodes **101**.

(50) AMF **1115** may include one or more devices, systems, Virtualized Network Functions (“VNFs”), Cloud-Native Network Functions (“CNFs”), etc., that perform operations to register UE **501** with the 5G network, to establish bearer channels associated with a session with UE **501**, to hand off UE **501** from the 5G network to another network, to hand off UE **501** from the other network to the 5G network, manage mobility of UE **501** between RANs **1110** and/or gNBs **1111**, and/or to perform other operations. In some embodiments, the 5G network may include multiple AMFs **1115**, which communicate with each other via the N14 interface (denoted in FIG. **11** by the line marked “N14” originating and terminating at AMF **1115**).

(51) MME **1116** may include one or more devices, systems, VNFs, CNFs, etc., that perform operations to register UE **501** with the EPC, to establish bearer channels associated with a session with UE **501**, to hand off UE **501** from the EPC to another network, to hand off UE **501** from another network to the EPC, manage mobility of UE **501** between RANs **1112** and/or eNBs **1113**, and/or to perform other operations.

(52) SGW **1117** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate traffic received from one or more eNBs **1113** and send the aggregated traffic to an external network or device via UPF/PGW-U **1135**. Additionally, SGW **1117** may aggregate traffic received from one or more UPF/PGW-Us **1135** and may send the aggregated traffic to one or more eNBs **1113**. SGW **1117** may operate as an anchor for the user plane during inter-eNB handovers and as an anchor for mobility between different telecommunication networks or RANs (e.g., RANs **1110** and **1112**).

(53) SMF/PGW-C **1120** may include one or more devices, systems, VNFs, CNFs, etc., that gather, process, store, and/or provide information in a manner described herein. SMF/PGW-C **1120** may, for example, facilitate the establishment of communication sessions on behalf of UE **501**. In some embodiments, the establishment of communications sessions may be performed in accordance with one or more policies provided by PCF/PCRF **1125**.

(54) PCF/PCRF **1125** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate information to and from the 5G network and/or other sources. PCF/PCRF **1125** may receive information regarding policies and/or subscriptions from one or more sources, such as subscriber databases and/or from one or more users (such as, for example, an administrator associated with PCF/PCRF **1125**).

(55) AF **1130** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide information that may be used in determining parameters (e.g., quality of service parameters, charging parameters, or the like) for certain applications.

(56) UPF/PGW-U **1135** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide data (e.g., user plane data). For example, UPF/PGW-U **1135** may receive user plane data (e.g., voice call traffic, data traffic, etc.), destined for UE **501**, from DN **1150**, and may forward the user plane data toward UE **501** (e.g., via RAN **1110**, SMF/PGW-C **1120**, and/or one or more other devices). In some embodiments, multiple UPFs **1135** may be deployed (e.g., in different geographical locations), and the delivery of content to UE **501** may be coordinated via the N9 interface (e.g., as denoted in FIG. **11** by the line marked “N9” originating and terminating at UPF/PGW-U **1135**). Similarly, UPF/PGW-U **1135** may receive traffic from UE **501** (e.g., via RAN **1110**, RAN **1112**, SMF/PGW-C **1120**, and/or one or more other devices), and may forward the traffic toward DN **1150**. In some embodiments, UPF/PGW-U **1135** may communicate (e.g., via the N4 interface) with SMF/PGW-C **1120**, regarding user plane data processed by UPF/PGW-U **1135**.

(57) UDM/HSS **1140** and AUSF **1145** may include one or more devices, systems, VNFs, CNFs, etc., that manage, update, and/or store, in one or more memory devices associated with AUSF **1145** and/or UDM/HSS **1140**, profile information associated with a subscriber. AUSF **1145** and/or UDM/HSS **1140** may perform authentication, authorization, and/or accounting operations associated with the subscriber and/or a communication session with UE **501**.

(58) DN **1150** may include one or more wired and/or wireless networks. For example, DN **1150** may include an Internet Protocol (“IP”)-based PDN, a wide area network (“WAN”) such as the Internet, a private enterprise network, and/or one or more other networks. UE **501** may communicate, through DN **1150**, with data servers, other UEs **501**, and/or to other servers or applications that are coupled to DN **1150**. DN **1150** may be connected to one or more other networks, such as a public switched telephone network (“PSTN”), a public land mobile network (“PLMN”), and/or another network. DN **1150** may be connected to one or more devices, such as content providers, applications, web servers, and/or other devices, with which UE **501** may communicate.

(59) FIG. **12** illustrates an example RAN environment **1200**, which may be included in and/or implemented by one or more RANs (e.g., RAN **1110**, RAN **1112**, or some other RAN). In some embodiments, a particular RAN may include one RAN environment **1200**. In some embodiments, a particular RAN may include multiple RAN environments **1200**. In some embodiments, RAN environment **1200** may correspond to a particular gNB **1111** of a 5G RAN (e.g., RAN **1110**). In some embodiments, RAN environment **1200** may correspond to multiple gNBs **1111**. In some embodiments, RAN environment **1200** may correspond to one or more other types of base stations of one or more other types of RANs. As shown, RAN environment **1200** may include Central Unit (“CU”) **1205**, one or more Distributed Units (“DUs”) **1203-1** through **1203-N** (referred to individually as “DU **1203**,” or collectively as “DUs **1203**”), and one or more Radio Units (“RUs”) **1201-1** through **1201-M** (referred to individually as “RU **1201**,” or collectively as “RUs **1201**”).

(60) CU **1205** may communicate with a core of a wireless network (e.g., may communicate with one or more of the devices or systems described above with respect to FIG. **11**, such as AMF **1115** and/or UPF/PGW-U **1135**). In the uplink direction (e.g., for traffic from UEs **501** to a core network), CU **1205** may aggregate traffic from DUs **1203**, and forward the aggregated traffic to the core network. In some embodiments, CU **1205** may receive traffic according to a given protocol (e.g., Radio Link Control (“RLC”)) from DUs **1203**, and may perform higher-layer processing (e.g., may aggregate/process RLC packets and generate Packet Data Convergence Protocol (“PDCP”) packets based on the RLC packets) on the traffic received from DUs **1203**.

(61) In accordance with some embodiments, CU **1205** may receive downlink traffic (e.g., traffic from the core network) for a particular UE **501**, and may determine which DU(s) **1203** should receive the downlink traffic. DU **1203** may include one or more devices that transmit traffic

between a core network (e.g., via CU **1205**) and UE **501** (e.g., via a respective RU **1201**). DU **1203** may, for example, receive traffic from RU **1201** at a first layer (e.g., physical (“PHY”) layer traffic, or lower PHY layer traffic), and may process/aggregate the traffic to a second layer (e.g., upper PHY and/or RLC). DU **1203** may receive traffic from CU **1205** at the second layer, may process the traffic to the first layer, and provide the processed traffic to a respective RU **1201** for transmission to UE **501**.

(62) RU **1201** may include hardware circuitry (e.g., one or more RF transceivers, antennas, radios, and/or other suitable hardware) to communicate wirelessly (e.g., via an RF interface) with one or more UEs **501**, one or more other DUs **1203** (e.g., via RUs **1201** associated with DUs **1203**), and/or any other suitable type of device. In the uplink direction, RU **1201** may receive traffic from UE **501** and/or another DU **1203** via the RF interface and may provide the traffic to DU **1203**. In the downlink direction, RU **1201** may receive traffic from DU **1203**, and may provide the traffic to UE **501** and/or another DU **1203**.

(63) One or more elements of RAN environment **1200** may, in some embodiments, be communicatively coupled to one or more Multi-Access/Mobile Edge Computing (“MEC”) devices, referred to sometimes herein simply as “MECs” **1207**. For example, DU **1203-1** may be communicatively coupled to MEC **1207-1**, DU **1203-N** may be communicatively coupled to MEC **1207-N**, CU **1205** may be communicatively coupled to MEC **1207-2**, and so on. MECs **1207** may include hardware resources (e.g., configurable or provisionable hardware resources) that may be configured to provide services and/or otherwise process traffic to and/or from UE **501**, via a respective RU **1201**.

(64) For example, DU **1203-1** may route some traffic, from UE **501**, to MEC **1207-1** instead of to a core network via CU **1205**. MEC **1207-1** may process the traffic, perform one or more computations based on the received traffic, and may provide traffic to UE **501** via RU **1201-1**. In some embodiments, MEC **1207** may include, and/or may implement, some or all of the functionality described above with respect to AF **1130**, UPF **1135**, and/or one or more other devices, systems, VNFs, CNFs, etc. In this manner, ultra-low latency services may be provided to UE **501**, as traffic does not need to traverse DU **1203**, CU **1205**, links between DU **1203** and CU **1205**, and an intervening backhaul network between RAN environment **1200** and the core network.

(65) FIG. **13** illustrates example components of device **1300**. One or more of the devices described above may include one or more devices **1300**. Device **1300** may include bus **1310**, processor **1320**, memory **1330**, input component **1340**, output component **1350**, and communication interface **1360**. In another implementation, device **1300** may include additional, fewer, different, or differently arranged components.

(66) Bus **1310** may include one or more communication paths that permit communication among the components of device **1300**. Processor **1320** may include a processor, microprocessor, or processing logic that may interpret and execute instructions. In some embodiments, processor **1320** may be or may include one or more hardware processors. Memory **1330** may include any type of dynamic storage device that may store information and instructions for execution by processor **1320**, and/or any type of non-volatile storage device that may store information for use by processor **1320**.

(67) Input component **1340** may include a mechanism that permits an operator to input information to device **1300** and/or other receives or detects input from a source external to input component **1340**, such as a touchpad, a touchscreen, a keyboard, a keypad, a button, a switch, a microphone or other audio input component, etc. In some embodiments, input component **1340** may include, or may be communicatively coupled to, one or more sensors, such as a motion sensor (e.g., which may be or may include a gyroscope, accelerometer, or the like), a location sensor (e.g., a Global Positioning System (“GPS”)-based location sensor or some other suitable type of location sensor or location determination component), a thermometer, a barometer, and/or some other type of sensor. Output component **1350** may include a mechanism that outputs information to the operator, such as

a display, a speaker, one or more light emitting diodes (“LEDs”), etc.

(68) Communication interface **1360** may include any transceiver-like mechanism that enables device **1300** to communicate with other devices and/or systems. For example, communication interface **1360** may include an Ethernet interface, an optical interface, a coaxial interface, or the like. Communication interface **1360** may include a wireless communication device, such as an infrared (“IR”) receiver, a Bluetooth® radio, or the like. The wireless communication device may be coupled to an external device, such as a remote control, a wireless keyboard, a mobile telephone, etc. In some embodiments, device **1300** may include more than one communication interface **1360**. For instance, device **1300** may include an optical interface and an Ethernet interface.

(69) Device **1300** may perform certain operations relating to one or more processes described above. Device **1300** may perform these operations in response to processor **1320** executing software instructions stored in a computer-readable medium, such as memory **1330**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **1330** from another computer-readable medium or from another device. The software instructions stored in memory **1330** may cause processor **1320** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(70) The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the possible implementations to the precise form disclosed. Modifications and variations are possible in light of the above disclosure or may be acquired from practice of the implementations.

(71) For example, while series of blocks and/or signals have been described above (e.g., with regard to FIGS. **1-10**), the order of the blocks and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel. Additionally, while the figures have been described in the context of particular devices performing particular acts, in practice, one or more other devices may perform some or all of these acts in lieu of, or in addition to, the above-mentioned devices.

(72) The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be designed based on the description herein.

(73) In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

(74) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of the possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

(75) Further, while certain connections or devices are shown, in practice, additional, fewer, or different, connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice, the functionality of multiple devices may be performed by a single device, or the functionality of one device may be performed by multiple devices. Further,

multiple ones of the illustrated networks may be included in a single network, or a particular network may include multiple networks. Further, while some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

(76) To the extent the aforementioned implementations collect, store, or employ personal information of individuals, groups or other entities, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage, and use of such information can be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as can be appropriate for the situation and type of information. Storage and use of personal information can be in an appropriately secure manner reflective of the type of information, for example, through various access control, encryption and anonymization techniques for particularly sensitive information.

(77) No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items, and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A device, comprising: one or more processors configured to: connect to a base station of a wireless network; receive, from a first Non-Terrestrial Network (“NTN”) node of a plurality of NTN nodes, telemetry information associated with the plurality of NTN nodes, wherein the received telemetry information is broadcasted from the first NTN node and is received by the device without connecting to the first NTN node, wherein the received telemetry information includes: first telemetry information associated with the first NTN node, and second telemetry information associated with a second NTN node of the plurality of NTN nodes; determine, based on the first telemetry information, that a distance between the device and the first NTN node is increasing as a function of time; determine, based on the second telemetry information, that a distance between the device and the second NTN node is decreasing as a function of time; identify a particular handover condition; select, based on identifying the particular handover condition, a particular NTN node of the plurality of NTN nodes, wherein selecting the particular NTN node includes selecting the second NTN node based on determining that the distance between the device and the first NTN node is increasing as a function of time and further based on determining that the distance between the device and the second NTN node is decreasing as a function of time; initiate a handover procedure from the base station of the wireless network to the second NTN node; and connect to the second NTN node based on the handover procedure.
2. The device of claim 1, wherein the plurality of NTN nodes are implemented by a plurality of satellites.
3. The device of claim 2, wherein the plurality of satellites each orbit the Earth.
4. The device of claim 2, wherein each NTN node, of the plurality of NTN nodes, is implemented by a respective particular satellite of the plurality of satellites.
5. The device of claim 1, wherein the telemetry information associated with the plurality of NTN nodes includes, for each NTN node, at least one of: a current position of the each NTN node, a timestamp associated with the current position of the each NTN node, a trajectory of the each NTN

node, or a timestamp associated with the trajectory of the each NTN node.

6. The device of claim 1, wherein the one or more processors are further configured to: determine a predicted route of the device, wherein determining that the distance between the device and the first NTN node is increasing as a function of time or that the distance between the device and the second NTN node is decreasing as a function of time is based on the predicted route of the device.

7. A non-transitory computer-readable medium, storing a plurality of processor-executable instructions to: connect to a base station of a wireless network; receive, from a first Non-Terrestrial Network (“NTN”) node of a plurality of NTN nodes, telemetry information associated with the plurality of NTN nodes, wherein the received telemetry information is broadcasted from the first NTN node and is received by the device without connecting to the first NTN node, wherein the received telemetry information includes: first telemetry information associated with the first NTN node, and second telemetry information associated with a second NTN node of the plurality of NTN nodes; determine, based on the first telemetry information, that a distance between a device and the first NTN node is increasing as a function of time; determine, based on the second telemetry information, that a distance between the device and the second NTN node is decreasing as a function of time; identify a particular handover condition; select, based on identifying the particular handover condition, a particular NTN node of the plurality of NTN nodes, wherein selecting the particular NTN node includes selecting the second NTN node based on determining that the distance between the device and the first NTN node is increasing as a function of time and further based on determining that the distance between the device and the second NTN node is decreasing as a function of time; initiate a handover procedure from the base station of the wireless network to the second NTN node; and connect to the second NTN node based on the handover procedure.

8. The non-transitory computer-readable medium of claim 7, wherein the plurality of NTN nodes are implemented by a plurality of satellites.

9. The non-transitory computer-readable medium of claim 8, wherein the plurality of satellites each orbit the Earth.

10. The non-transitory computer-readable medium of claim 8, wherein each NTN node, of the plurality of NTN nodes, is implemented by a respective particular satellite of the plurality of satellites.

11. The non-transitory computer-readable medium of claim 7, wherein the telemetry information associated with the plurality of NTN nodes includes, for each NTN node, at least one of: a current position of the each NTN node, a timestamp associated with the current position of the each NTN node, a trajectory of the each NTN node, or a timestamp associated with the trajectory of the each NTN node.

12. The non-transitory computer-readable medium of claim 7, wherein the plurality of processor-executable instructions further include processor-executable instructions to: determine a predicted route of the device, wherein determining that the distance between the device and the first NTN node is increasing as a function of time or that the distance between the device and the second NTN node is decreasing as a function of time is based on the predicted route of the device.

13. A method, comprising: connecting to a base station of a wireless network; receiving, from a first Non-Terrestrial Network (“NTN”) node of a plurality of NTN nodes, telemetry information associated with the plurality of NTN nodes, wherein the received telemetry information is broadcasted from the first NTN node and is received by the device without connecting to the first NTN node, wherein the received telemetry information includes: first telemetry information associated with the first NTN node, and second telemetry information associated with a second NTN node of the plurality of NTN nodes; determining, based on the first telemetry information, that a distance between a device and the first NTN node is increasing as a function of time; determining, based on the second telemetry information, that a distance between the device and the second NTN node is decreasing as a function of time; identifying a particular handover condition;

selecting, based on identifying the particular handover condition, a particular NTN node of the plurality of NTN nodes, wherein selecting the particular NTN node includes selecting the second NTN node based on determining that the distance between the device and the first NTN node is increasing as a function of time and further based on determining that the distance between the device and the second NTN node is decreasing as a function of time; initiating a handover procedure from the base station of the wireless network to the second NTN node; and connecting to the second NTN node based on the handover procedure.

14. The method of claim 13, wherein the plurality of NTN nodes are implemented by a plurality of satellites.

15. The method of claim 14, wherein the plurality of satellites each orbit the Earth.

16. The method of claim 14, wherein each NTN node, of the plurality of NTN nodes, is implemented by a respective particular satellite of the plurality of satellites.

17. The method of claim 13, wherein the telemetry information associated with the plurality of NTN nodes includes, for each NTN node, at least one of: a current position of the each NTN node, a timestamp associated with the current position of the each NTN node, a trajectory of the each NTN node, or a timestamp associated with the trajectory of the each NTN node.

18. The method of claim 13, further comprising: determining a predicted route of the device, wherein determining that the distance between the device and the first NTN node is increasing as a function of time or that the distance between the device and the second NTN node is decreasing as a function of time is based on the predicted route of the device.
